

EN.553.724 Probabilistic Machine Learning

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Chapter 1

Introduction

1.1 Probabilistic Modeling and Learning

1.1.1 Classical vs. Modern Methods

Classical probabilistic models (e.g. graphical models) and modern deep generative models differ primarily in how structure is specified.

- **Classical approaches**

- Leverage domain knowledge to specify model structure.
- Work well when the model is correctly specified.
- Provide interpretable probabilistic quantities and uncertainty quantification.

- **Modern approaches**

- Learn highly complex distributions directly from data.
- Require large datasets.
- Often lack interpretability.

- **Goal**

combine structured probabilistic modeling with flexible function approximation.

1.1.2 Learning a Function

Model:

$$y = f(x) + \varepsilon, \quad \mathbb{E}[\varepsilon|x] = 0.$$

Given data $\{(x_i, y_i)\}_{i=1}^N$, choose f_θ .

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N L(f_\theta(x_i), y_i).$$

1.1.3 Maximum Likelihood Interpretation

Assume:

$$\varepsilon \sim \mathcal{N}(0, I).$$

Then:

$$p(y_i|x_i, \theta) \propto \exp\left(-\frac{1}{2}\|y_i - f_\theta(x_i)\|^2\right).$$

Proposition 1.1.1. *Minimizing squared loss is equivalent to maximum likelihood estimation.*

$$\arg \max_{\theta} p(y_{1:N}|x_{1:N}) = \arg \min_{\theta} \sum_{i=1}^N \|y_i - f_\theta(x_i)\|^2.$$

1.1.4 Learning a Distribution

Given $x_1, \dots, x_N \stackrel{\text{iid}}{\sim} \mathbb{P}$:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^N \log p_\theta(x_i) = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N -\log p_\theta(x_i).$$

1.2 Measure-Theoretic Foundations

1.2.1 Discrete and Continuous Probability

- Discrete:

$$\mathbb{P}(A) = \sum_{x \in A} p(x)$$

- Continuous:

$$\mathbb{P}(A) = \int_A p(x) \, dx$$

Unified via:

$$\mathbb{P}(A) = \int_A p(x) \, \lambda(\mathrm{d}x).$$

1.2.2 Probability Spaces

Definition 1.2.1. A measurable space is $(\Omega, \mathcal{F})$.

Definition 1.2.2. A probability measure $\mathbb{P}$ satisfies:

- $\mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- countable additivity

If a density exists:

$$\mathbb{P}(A) = \int_A p(x) \, \lambda(\mathrm{d}x).$$

1.2.3 Expectation

$$\mathbb{E}[f] = \int f(x) \mathbb{P}(\mathrm{d}x)$$

If density exists:

$$\mathbb{E}[f] = \int f(x)p(x) \lambda(\mathrm{d}x).$$

1.2.4 Radon–Nikodym Derivative

Definition 1.2.3. $\mathbb{P} \ll Q$ if $Q(A) = 0 \Rightarrow \mathbb{P}(A) = 0$.

Theorem 1.2.1 (Radon–Nikodym). *If $\mathbb{P} \ll Q$, then*

$$\int f \mathrm{d}\mathbb{P} = \int f \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q} \mathrm{d}Q.$$

If densities exist:

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}Q}(x) = \frac{p(x)}{q(x)}.$$

1.2.5 Random Variables

Definition 1.2.4. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$.

Distribution:

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A).$$

1.2.6 Conditional Probability

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

$$p(x|y) = \frac{p(x)p(y|x)}{p(y)}$$

1.2.7 Beta-Binomial Model

$$\theta \sim \text{Beta}(\alpha, \beta), \quad X|\theta \sim \text{Binomial}(m+n, \theta).$$

Posterior:

$$\theta|X \sim \text{Beta}(\alpha+m, \beta+n).$$

1.2.8 Conditional Independence

Definition 1.2.5. $X \perp Y|Z$ if

$$p(x, y|z) = p(x|z)p(y|z).$$

1.3 Divergences and Information

1.3.1 Total Variation

Definition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \frac{1}{2} \int |p - q| \lambda(\mathrm{d}x).$$

Proposition 1.3.1.

$$\text{TV}(\mathbb{P}, Q) = \max_A |\mathbb{P}(A) - Q(A)|.$$

1.3.2 f -Divergence

Definition 1.3.2.

$$D_f(\mathbb{P}||Q) = \mathbb{E}_Q \left[f \left(\frac{p}{q} \right) \right].$$

$$\text{KL}(\mathbb{P}||Q) = \int p \log \frac{p}{q}.$$

1.3.3 Non-Negativity

Theorem 1.3.1 (Jensen).

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Proposition 1.3.2.

$$D_f(\mathbb{P}||Q) \geq 0.$$

Proof.

$$\mathbb{E}_Q f \left(\frac{p}{q} \right) \geq f \left(\mathbb{E}_Q \frac{p}{q} \right) = f(1) = 0.$$

□

1.3.4 Entropy and KL

Definition 1.3.3.

$$H(\mathbb{P}) = - \int p \log p.$$

$$\text{KL}(\mathbb{P}||Q) = -H(\mathbb{P}) - \mathbb{E}_{\mathbb{P}}[\log q].$$

1.3.5 MLE and KL

Proposition 1.3.3.

$$\mathbb{E}_{\mathbb{P}}[-\log p_{\theta}(x)] = \text{KL}(\mathbb{P} \parallel \mathbb{P}_{\theta}) + H(\mathbb{P}).$$

$$\Rightarrow \text{MLE} = \arg \min_{\theta} \text{KL}(\mathbb{P} \parallel \mathbb{P}_{\theta}).$$

1.3.6 KL Direction

$$\text{KL}(\mathbb{P} \parallel Q) \quad \text{mass-covering}$$

$$\text{KL}(Q \parallel \mathbb{P}) \quad \text{mode-seeking}$$

1.3.7 Invariance

Proposition 1.3.4. *For invertible g :*

$$D_f(\mathbb{P} \parallel Q) = D_f(g_{\#}\mathbb{P} \parallel g_{\#}Q).$$

1.4 Optimal Transport and Integral Probability Metrics

1.4.1 Couplings and Wasserstein Distance

Definition 1.4.1. Let $(\Omega, \mathcal{F})$ be a measurable space with metric d , and let μ, ν be probability measures on Ω .

A *coupling* of μ and ν is a probability measure γ on $\Omega \times \Omega$ such that:

$$\gamma(A \times \Omega) = \mu(A), \quad \gamma(\Omega \times A) = \nu(A).$$

Interpretation: a coupling specifies a joint distribution (X, Y) with marginals $X \sim \mu$, $Y \sim \nu$.

Definition 1.4.2. The Wasserstein- p distance is defined as

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \mathbb{E}_{(X, Y) \sim \gamma} [d(X, Y)^p] \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ, ν .

Remark 1.4.1. W_1 is often called the *earth mover's distance*, since it measures the minimal cost of transporting mass from μ to ν .

1.4.2 Integral Probability Metrics

Definition 1.4.3. Let $\mathcal{F}$ be a class of functions $f : \Omega \rightarrow \mathbb{R}$. The integral probability metric (IPM) is

$$d_{\mathcal{F}}(\nu, \mu) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f]|.$$

Lemma 1.4.1. Let $\mathcal{F} = \{f : \|f\|_{\infty} \leq 1\}$. Then

$$\text{TV}(\mu, \nu) = \frac{1}{2} d_{\mathcal{F}}(\nu, \mu).$$

Proof. For any f with $\|f\|_{\infty} \leq 1$,

$$|\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f]| = \left| \int f(x)(\nu - \mu)(dx) \right| \leq \int |f(x)| |\nu - \mu|(dx) \leq \int |\nu - \mu|(dx).$$

Taking the supremum gives

$$d_{\mathcal{F}}(\nu, \mu) \leq 2 \text{TV}(\mu, \nu).$$

Equality is achieved by choosing

$$f(x) = \text{sign}(p(x) - q(x)),$$

which yields

$$\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f] = \int |p(x) - q(x)| dx.$$

□

1.4.3 Kantorovich–Rubinstein Duality

Theorem 1.4.1 (Kantorovich–Rubinstein). Let $\mathcal{F}$ be the set of 1-Lipschitz functions on $\mathbb{R}^d$. Then

$$W_1(\mu, \nu) = d_{\mathcal{F}}(\nu, \mu).$$

Sketch. For any coupling γ and any 1-Lipschitz f ,

$$|f(x) - f(y)| \leq d(x, y).$$

Thus

$$\mathbb{E}_{\nu}[f] - \mathbb{E}_{\mu}[f] = \mathbb{E}_{\gamma}[f(Y) - f(X)] \leq \mathbb{E}_{\gamma}[d(X, Y)].$$

Taking infimum over γ gives

$$d_{\mathcal{F}}(\nu, \mu) \leq W_1(\mu, \nu).$$

The reverse inequality follows by constructing an optimal dual potential. □

Remarks on Metrics vs Divergences

- Total variation and Wasserstein are *metrics*.
- f -divergences (e.g. KL) are not metrics:
 - not symmetric
 - do not satisfy triangle inequality
- IPMs provide a unifying view of distances via function classes.

1.5 Maximum Entropy and Exponential Families

1.5.1 Maximum Entropy Principle

Let P be an unknown distribution on Ω . Suppose we are given moment constraints:

$$\mathbb{E}_P[\phi_i(X)] = \mu_i, \quad i = 1, \dots, d,$$

or equivalently

$$\mathbb{E}_P[\phi(X)] = \mu \in \mathbb{R}^d,$$

where $\phi : \Omega \rightarrow \mathbb{R}^d$.

Remark 1.5.1. These constraints alone do not uniquely determine P .

A natural principle is to choose the *least informative* distribution satisfying the constraints.

Definition 1.5.1. The maximum entropy distribution (with respect to a reference measure ν) is

$$\arg \max_P H_\nu(P) \quad \text{s.t.} \quad \mathbb{E}_P[\phi] = \mu,$$

where

$$H_\nu(P) = - \int p(x) \log p(x) \nu(\mathrm{d}x).$$

Derivation via Lagrange Multipliers

Assume Ω is finite for simplicity, and write $p(x) = P(X = x)$.

We maximize

$$- \sum_{x \in \Omega} p(x) \log p(x)$$

subject to:

$$\sum_x p(x) = 1, \quad \sum_x p(x) \phi_i(x) = \mu_i.$$

Define the Lagrangian:

$$\mathcal{L}(p, \lambda_0, \lambda) = - \sum_x p(x) \log p(x) + \lambda_0 \left(\sum_x p(x) - 1 \right) + \sum_{i=1}^d \lambda_i \left(\sum_x p(x) \phi_i(x) - \mu_i \right).$$

Taking derivatives:

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -\log p(x) - 1 + \lambda_0 + \sum_{i=1}^d \lambda_i \phi_i(x).$$

Setting to zero:

$$\log p(x) = \lambda_0 - 1 + \sum_{i=1}^d \lambda_i \phi_i(x).$$

Thus

$$p(x) \propto \exp\left(\sum_{i=1}^d \lambda_i \phi_i(x)\right) = \exp(\langle \theta, \phi(x) \rangle),$$

where $\theta = (\lambda_1, \dots, \lambda_d)$.

This gives the exponential family form.

1.5.2 Exponential Family

Definition 1.5.2. Let $(\mathcal{X}, \nu)$ be a measure space and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$.

The exponential family is

$$p_\theta(x) = \frac{\exp(\langle \theta, \phi(x) \rangle)}{Z(\theta)} = \exp(\langle \theta, \phi(x) \rangle - A(\theta)),$$

where

$$Z(\theta) = \int \exp(\langle \theta, \phi(x) \rangle) \nu(\mathrm{d}x), \quad A(\theta) = \log Z(\theta).$$

Remark 1.5.2. • θ is the natural parameter

- $\phi(x)$ are sufficient statistics
- $A(\theta)$ is the log-partition function

1.5.2.1 Example: Gaussian from Exponential Family

Let $\phi(x) = (x, x^2)$ on $\mathbb{R}$.

Then

$$p_\theta(x) \propto \exp(\theta_1 x + \theta_2 x^2).$$

Completing the square:

$$\theta_1 x + \theta_2 x^2 = \theta_2 \left(x + \frac{\theta_1}{2\theta_2}\right)^2 - \frac{\theta_1^2}{4\theta_2}.$$

For $\theta_2 < 0$, this is a Gaussian:

$$p_\theta(x) = \mathcal{N}(\mu, \sigma^2), \quad \mu = -\frac{\theta_1}{2\theta_2}, \quad \sigma^2 = -\frac{1}{2\theta_2}.$$

1.5.3 Maximum Entropy Characterization

Theorem 1.5.1. *If there exists θ such that*

$$\mathbb{E}_{p_\theta}[\phi(X)] = \mu,$$

then p_θ is the maximum entropy distribution among all distributions satisfying the moment constraints.

Proof. Let P satisfy $\mathbb{E}_P[\phi] = \mu$. Then

$$\text{KL}(P||p_\theta) = \mathbb{E}_P[\log p(x) - \log p_\theta(x)] \geq 0.$$

Rearranging:

$$-H(P) \geq \mathbb{E}_P[\log p_\theta(x)].$$

Using exponential family form:

$$\mathbb{E}_P[\log p_\theta(x)] = \mathbb{E}_P[\langle \theta, \phi(x) \rangle - A(\theta)] = \langle \theta, \mu \rangle - A(\theta).$$

Thus

$$H(P) \leq A(\theta) - \langle \theta, \mu \rangle = H(p_\theta),$$

so p_θ maximizes entropy. □

1.5.4 Mean Parameter Mapping

Define the mapping

$$\nabla A(\theta) = \mathbb{E}_{p_\theta}[\phi(X)].$$

Definition 1.5.3. Let

$$\mathcal{M} = \{\mu \in \mathbb{R}^d : \exists P \text{ such that } \mathbb{E}_P[\phi(X)] = \mu\}$$

be the set of achievable mean parameters.

Definition 1.5.4. The statistics $\{\phi_i\}_{i=1}^d$ are *minimal* if

$$\{1, \phi_1, \dots, \phi_d\} \text{ are linearly independent.}$$

Proposition 1.5.1. *If ϕ is minimal, then:*

- $\nabla A : \Theta \rightarrow \mathcal{M}$ is injective
- $\nabla A : \Theta \rightarrow \mathcal{M}^\circ$ is surjective (onto the interior)

Sketch. If ϕ is minimal, then $A(\theta)$ is strictly convex:

$$\nabla^2 A(\theta) = \text{Cov}_{p_\theta}(\phi(X)) \succ 0.$$

Strict convexity implies injectivity of ∇A .

Surjectivity onto the interior follows from convex duality and properties of cumulant generating functions. □

1.5.5 Convexity and Geometry

Proposition 1.5.2. *The log-partition function satisfies:*

$$\nabla^2 A(\theta) = \text{Cov}_{p_\theta}(\phi(X)) \succeq 0.$$

Remark 1.5.3. • $A(\theta)$ is convex

- strictly convex if ϕ is minimal
- exponential families form a convex dual pair:

$$\theta \leftrightarrow \mu = \nabla A(\theta)$$

Non-Minimal Case

If ϕ is not minimal, then there exists $v \neq 0$ such that

$$\langle v, \phi(x) \rangle = c \quad \forall x.$$

Then

$$p_{\theta+tv}(x) = p_\theta(x),$$

so the parametrization is not identifiable, and ∇A is not injective.

Chapter 2

Graphical Models

2.1 Directed Graphical Models (Bayesian Networks)

2.1.1 Factorization in DAGs

Definition 2.1.1. Let $G = (V, E)$ be a directed acyclic graph (DAG), and let $\{X_v\}_{v \in V}$ be random variables.

A distribution P *factorizes according to* G if

$$p(x_V) = \prod_{v \in V} p(x_v \mid x_{\text{pa}(v)}),$$

where $\text{pa}(v)$ denotes the set of parents of node v .

Remark 2.1.1. Each factor $p(x_v \mid x_{\text{pa}(v)})$ is a *local conditional model*.

Definition 2.1.2. A *topological ordering* of a DAG is an ordering $(v_1, \dots, v_n)$ such that

$$v_i \rightarrow v_j \implies i < j.$$

Remark 2.1.2. This allows a generative interpretation:

$$X_{v_1} \rightarrow X_{v_2} \rightarrow \dots \rightarrow X_{v_n},$$

sampling each variable conditioned on its parents.

2.1.2 Independence Structure

Definition 2.1.3. Let $S_v = (\{v\} \cup \text{Desc}(v) \cup \text{pa}(v))^c$.

A DAG encodes the independence relations

$$X_v \perp X_{S_v} \mid X_{\text{pa}(v)}.$$

Proposition 2.1.1. *A distribution factorizes according to G if and only if it satisfies the independence relations encoded by G .*

2.1.3 Factorization $\Rightarrow$ Independence (Key Idea)

Lemma 2.1.1. *If P factorizes according to a DAG G , then for any upward-closed set U ,*

$$p(x_U) = \prod_{u \in U} p(x_u \mid x_{\text{pa}(u)}).$$

Sketch. Sum out variables not in U using the factorization; terms outside U cancel due to normalization. $\square$

This leads to conditional independence:

$$X_v \perp X_{S_v} \mid X_{\text{pa}(v)}.$$

2.1.4 Constructing a DAG

Given variables $X_1, \dots, X_n$:

- Choose an ordering $X_1, \dots, X_n$
- For each k , choose a minimal set $S \subseteq \{1, \dots, k-1\}$ such that

$$X_k \perp X_{[k-1] \setminus S} \mid X_S$$

- Set $\text{pa}(k) = S$

Remark 2.1.3. The resulting graph is not unique.

Remark 2.1.4. Causality can guide graph construction, but cannot generally be inferred from the graph alone.

2.1.5 Compactness of Representation

For n variables taking values in $\{1, \dots, r\}$:

- Full joint distribution requires:

$$r^n - 1 \quad \text{parameters}$$

- If max indegree is d :

$$O(nr^{d+1})$$

parameters suffice

Remark 2.1.5. Graphical models exploit conditional independence for exponential savings.

2.1.6 Examples

2.1.6.1 Markov Chain

$$X_{k+1} \perp X_1, \dots, X_{k-1} \mid X_k$$

Factorization:

$$p(x_1, \dots, x_n) = p(x_1) \prod_{k=2}^n p(x_k \mid x_{k-1})$$

2.1.6.2 d -th Order Markov Model

$$X_{k+d} \perp X_1, \dots, X_{k-1} \mid X_k, \dots, X_{k+d-1}$$

2.1.6.3 Naive Bayes

Latent class C :

$$X_k \perp X_{-k} \mid C$$

Factorization:

$$p(c, x_1, \dots, x_n) = p(c) \prod_{k=1}^n p(x_k \mid c)$$

2.1.7 Latent Variable Models

Definition 2.1.4. A *latent variable model* introduces hidden variables Z such that

$$p(x) = \int p(x, z) \, dz.$$

Remark 2.1.6. Latent variables simplify structure by introducing conditional independence.

2.1.7.1 Mixture Model

$$z_i \sim \text{Multinomial}(\pi), \quad x_i \mid z_i \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$$

2.1.7.2 Stochastic Block Model

$$z_i \sim \text{Multinomial}(\pi), \quad A_{ij} \mid z_i, z_j \sim \text{Bernoulli}(p_{z_i z_j})$$

2.1.7.3 Noisy-OR Model

$$x_j \mid z \sim \text{Bernoulli} \left(1 - \exp \left(- \sum_i w_{ij} z_i \right) \right)$$

2.1.7.4 Hidden Markov Model (HMM)

$$z_t = Az_{t-1} + \epsilon_t, \quad x_t = Cz_t + \eta_t$$

Remark 2.1.7. This is a linear Gaussian state-space model.

2.1.8 d-Separation

Definition 2.1.5. A path in a DAG is *blocked* by a set C if one of the following holds:

1. Chain: $X \rightarrow Z \rightarrow Y$ or $X \leftarrow Z \leftarrow Y$, and $Z \in C$
2. Fork: $X \leftarrow Z \rightarrow Y$, and $Z \in C$
3. Collider: $X \rightarrow Z \leftarrow Y$, and $Z \notin C$ and no descendant of Z is in C

Definition 2.1.6. A and B are d -separated by C if all paths between them are blocked.

Theorem 2.1.1. If P factorizes according to G and A, B are d -separated by C , then

$$X_A \perp X_B \mid X_C.$$

2.2 Undirected Graphical Models (Markov Random Fields)

2.2.1 Factorization

Definition 2.2.1. Let $G = (V, E)$ be an undirected graph. Let $\mathcal{C}$ be the set of maximal cliques.

A distribution factorizes over G if

$$p(x) \propto \exp(-E(x)), \quad E(x) = \sum_{C \in \mathcal{C}} \phi_C(x_C),$$

or equivalently

$$p(x) = \prod_{C \in \mathcal{C}} \psi_C(x_C).$$

Remark 2.2.1. $\psi_C = e^{-\phi_C}$ are called *clique potentials*.

2.2.2 Markov Properties

Definition 2.2.2. P satisfies:

- Global Markov property:

$$X_A \perp X_B \mid X_C \quad \text{if } C \text{ separates } A, B$$

- Local Markov property:

$$X_v \perp X_{V \setminus (\{v\} \cup N(v))} \mid X_{N(v)}$$

- Pairwise Markov property:

$$X_v \perp X_w \mid X_{V \setminus \{v, w\}}, \quad (v, w) \notin E$$

Remark 2.2.2. Under positivity conditions:

$$\text{global} \iff \text{local} \iff \text{pairwise}$$

2.2.3 Energy-Based Models

Definition 2.2.3. An energy-based model defines

$$p_\theta(x) \propto e^{-E_\theta(x)}.$$

Remark 2.2.3. • Lower energy $\Rightarrow$ higher probability

- $E_\theta(x) = \langle \theta, \phi(x) \rangle$ recovers exponential families

2.2.4 Example: Ising Model

$$p(x) \propto \exp \left(\sum_i h_i x_i + \sum_{(i,j) \in E} J_{ij} x_i x_j \right), \quad x_i \in \{-1, 1\}$$

Conditional distribution:

$$p(x_i = 1 \mid x_{-i}) = \frac{\exp(h_i + \sum_{j \in N(i)} J_{ij} x_j)}{\exp(h_i + \sum_{j \in N(i)} J_{ij} x_j) + \exp(-h_i - \sum_{j \in N(i)} J_{ij} x_j)}.$$

Remark 2.2.4. Unlike DAGs, sampling is generally difficult due to the global partition function.

2.3 Additional Properties of Independence

Intersection Property

Proposition 2.3.1. *Suppose $p(x, y, z, w)$ has positive density with respect to a product measure. Then the intersection property holds:*

$$\begin{cases} X \perp Y \mid Z, W, \\ X \perp W \mid Z, Y \end{cases} \Rightarrow X \perp (Y, W) \mid Z.$$

Remark 2.3.1. This property does *not* hold in general without positivity.

Proposition 2.3.2. *If the intersection property holds, then the global, local, and pairwise Markov properties are equivalent.*

Hammersley–Clifford Theorem

Theorem 2.3.1 (Hammersley–Clifford). *Let $G = (V, E)$ be an undirected graph. Suppose P has positive density.*

Then:

$$P \text{ satisfies the Markov properties} \iff P \text{ factorizes over } G.$$

Remark 2.3.2. Positivity is essential: without it, independence does not imply factorization.

Canonical Representation

Theorem 2.3.2. *Any strictly positive distribution can be written as*

$$p(x) = \exp \left(\sum_{S \subseteq V} \xi_S(x_S) \right),$$

where

$$\xi_S(x_S) = \sum_{Z \subseteq S} (-1)^{|S|-|Z|} \log p(x_Z, x_{S^c} = 0).$$

Remark 2.3.3. • This is an inclusion–exclusion type expansion

- If P factorizes over G , then $\xi_S = 0$ for non-cliques S

2.4 Factor Graphs

Definition 2.4.1. A *factor graph* is a bipartite graph with:

- variable nodes V
- factor nodes F

and edges only between variables and factors.

A distribution factorizes as

$$p(x_V) = \prod_{a \in F} \varphi_a(x_{N(a)}),$$

where $N(a)$ are variables connected to factor a .

Remark 2.4.1. Factor graphs unify:

- DAG factorizations
- MRF factorizations

2.4.1 Example: Restricted Boltzmann Machine (RBM)

Let visible units $v \in \{0, 1\}^{d_v}$ and hidden units $h \in \{0, 1\}^{d_h}$.

$$p(v, h) \propto \exp(\langle v, Wh \rangle + \langle b, v \rangle + \langle a, h \rangle)$$

Remark 2.4.2. Key property:

$$p(h | v) = \prod_j p(h_j | v), \quad p(v | h) = \prod_i p(v_i | h)$$

Remark 2.4.3. This conditional independence enables efficient Gibbs sampling.

2.4.2 Example: Coding Theory (LDPC)

$$p(x, z) \propto \prod_i \mathbb{1}\{H_i z = 0\} \prod_j p(x_j | z_j)$$

Remark 2.4.4. Inference corresponds to belief propagation.

2.5 Conditional Random Fields

Definition 2.5.1. Let $(V \cup W, E)$ be an undirected graph.

A *conditional random field (CRF)* is a conditional distribution

$$P(X_W | X_V) \propto \prod_{C \in \mathcal{C}} \psi_C(x_C),$$

where cliques C involve variables in W .

Remark 2.5.1. CRFs model conditional structure without specifying a joint model over (V, W) .

2.6 Directed $\rightarrow$ Undirected Conversion

Definition 2.6.1. The *moralized graph* $\mathcal{M}[G]$ of a DAG G is obtained by:

- adding edges between all pairs of parents of each node
- dropping edge directions

Proposition 2.6.1. *If P factorizes according to a DAG G , then it factorizes according to the undirected graph $\mathcal{M}[G]$.*

2.7 Independence via Moralization

Theorem 2.7.1. *Let G be a DAG and A, B, C disjoint sets. The following are equivalent:*

1. *A and B are d -separated by C in G*
2. *A and B are separated by C in the moralized ancestral graph*

Moreover, if P factorizes according to G , then

$$X_A \perp X_B \mid X_C.$$

Definition 2.7.1. The *ancestral graph* is the subgraph induced by A, B, C and their ancestors.

2.8 Markov Blanket

Definition 2.8.1. A set U is a *Markov blanket* for v if

$$X_v \perp X_{V \setminus (U \cup \{v\})} \mid X_U.$$

Proposition 2.8.1. 1. *In an undirected graph, $N(v)$ is a Markov blanket.*

2. *In a DAG, the Markov blanket is:*

$$\text{pa}(v) \cup \text{ch}(v) \cup \text{co-pa}(v).$$

Remark 2.8.1. The Markov blanket gives the minimal conditioning set needed to isolate a node.

2.9 Learning Graph Structure (Undirected Case)

Proposition 2.9.1. *Assume P is positive. Define graph G by:*

$$(v, w) \in E \iff X_v \not\perp X_w \mid X_{V \setminus \{v, w\}}.$$

Then G is the unique minimal graph for which P satisfies the pairwise Markov property.

Comparison: Directed vs Undirected Models

- Directed models:
 - capture causal / generative structure
 - sampling is straightforward
 - require ordering / domain knowledge
- Undirected models:
 - symmetric relationships
 - easier structure learning
 - inference often harder (partition function)

2.10 Inference Problems

2.10.1 Core Tasks

Given a probabilistic model $p_\theta(x)$:

- **Sampling:**

$$x \sim p_\theta(x)$$

- **Marginals:**

$$p_\theta(x_i) = \int p_\theta(x) \nu^{\otimes(n-1)}(dx_{-i})$$

- **MAP (maximum a posteriori):**

$$x^* = \arg \max_x p_\theta(x)$$

- **Partition function:**

$$Z(\theta) = \int \psi_\theta(x) \nu^{\otimes n}(dx)$$

Remark 2.10.1. These tasks are tightly coupled: computing one often requires solving the others.

Latent Variable Models

For joint models:

$$p_\theta(x, z) \propto \psi_\theta(x, z)$$

posterior inference requires:

$$p_\theta(z|x) = \frac{\psi_\theta(x, z)}{Z(\theta, x)}, \quad Z(\theta, x) = \int \psi_\theta(x, z) \nu_Z(dz)$$

- Sampling: $z \sim p_\theta(z|x)$
- Marginals: $p_\theta(z_i|x)$
- MAP: $\arg \max_z p_\theta(z|x)$

2.11 Computational Complexity

Complexity Classes

- $O(n)$, $O(n \log n)$, $O(n^c)$: polynomial time
- $O(2^n)$: exponential time

Definition 2.11.1. A *decision problem* outputs YES/NO.

- **P**: solvable in polynomial time
- **NP**: solutions verifiable in polynomial time
- **NP-complete**: hardest problems in NP
- **NP-hard**: at least as hard as NP-complete

Example 2.11.1. 3-SAT: given a Boolean formula, does a satisfying assignment exist?

Counting Problems

Definition 2.11.2. #P: counting versions of NP problems (e.g., number of satisfying assignments).

Remark 2.11.1. Computing partition functions and marginals is typically #P-hard.

Theorem 2.11.1. *Computing partition functions or marginals in general graphical models is #P-hard.*

Remark 2.11.2. Even approximate counting (e.g., #SAT) can be NP-hard.

2.12 Approximation Algorithms

Definition 2.12.1. A *polynomial-time randomized approximation scheme (PRAS)* for Z outputs $\hat{Z}$ such that

$$e^{-\epsilon} Z \leq \hat{Z} \leq e^{\epsilon} Z$$

in polynomial time, with high probability.

Definition 2.12.2. • **FPRAS**: runtime polynomial in n and $1/\epsilon$

- **PTAS**: deterministic approximation scheme

Equivalence: Sampling, Marginals, Partition Function

Definition 2.12.3. A family of problems is *self-reducible* if fixing variables reduces the problem to a smaller instance.

Theorem 2.12.1 (informal). *For distributions $p_{\theta}(x) \propto \psi_{\theta}(x)$ on finite spaces:*

1. *Approximate sampling $\Rightarrow$ approximate marginals*
2. *Approximate marginals $\Rightarrow$ approximate sampling (if self-reducible)*
3. *Marginals $\Leftrightarrow$ partition function*

Theorem 2.12.2. *If we can sample within TV distance ϵ in $\text{poly}(n, 1/\epsilon)$ time, then we can approximate marginals via empirical averages:*

$$\hat{p}(x_i = a) = \frac{1}{N} \sum_{k=1}^N \mathbb{1}\{x_i^{(k)} = a\}$$

Proof. By Hoeffding's inequality:

$$\mathbb{P}(|\hat{p} - p| \geq \delta) \leq 2 \exp(-2N\delta^2)$$

so $N = O(1/\delta^2)$ suffices. □

2.13 Exact Inference

2.13.1 Variable Elimination

Algorithm 1 Variable Elimination

Input: $p(x) \propto \prod_a \psi_a(x_{\partial a})$
 Choose elimination order
for variable i **do**
 Multiply all factors involving i
 Sum out i
 Introduce new factor
end for
 Output desired marginal

Remark 2.13.1. Intermediate factors grow in size $\rightarrow$ exponential complexity.

Treewidth

Definition 2.13.1. The *treewidth* $\text{tw}(G)$ is the size of the largest intermediate factor (minus one) in optimal elimination ordering.

Remark 2.13.2. Complexity of variable elimination:

$$O(n \cdot \text{state}^{\text{tw}(G)+1})$$

2.14 Belief Propagation

2.14.1 Message Passing

Messages:

- Variable $\rightarrow$ factor:

$$m_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} m_{b \rightarrow i}(x_i)$$

- Factor $\rightarrow$ variable:

$$m_{a \rightarrow i}(x_i) \propto \sum_{x_{\partial a \setminus i}} \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} m_{j \rightarrow a}(x_j)$$

Marginals

$$p(x_i) \propto \prod_{a \in \partial i} m_{a \rightarrow i}(x_i)$$

$$p(x_{\partial a}) \propto \psi_a(x_{\partial a}) \prod_{j \in \partial a} m_{j \rightarrow a}(x_j)$$

2.14.2 Properties

- Exact on trees
- Equivalent to dynamic programming
- Linear-time in number of nodes (for trees)

Remark 2.14.1. On loopy graphs:

- becomes *loopy belief propagation*
- no guarantees, but often works well

2.15 Extensions

- Conditioning: add indicator factors
- Continuous variables: approximate messages (e.g., Gaussian)
- Junction tree algorithm: exact inference via tree decomposition

2.16 Takeaways

- Exact inference is exponential in general
- Tree structure enables efficient computation
- Message passing = dynamic programming on graphs
- Sampling, marginals, and partition function are deeply connected

Correctness of Belief Propagation (Trees)

We formalize what the messages compute.

Let $T_{b \rightarrow i}$ denote the subtree rooted at b when edge (b, i) is removed.

Proposition 2.16.1. *For a tree-structured factor graph:*

$$Z_{b \rightarrow i}(x_i) = \sum_{x_{V(T_{b \rightarrow i}) \setminus i}} \prod_{c \in F(T_{b \rightarrow i})} \psi_c(x_{\partial c}),$$

and similarly

$$Z_{a \rightarrow j}(x_j) = \sum_{x_{V(T_{a \rightarrow j}) \setminus j}} \prod_{c \in F(T_{a \rightarrow j})} \psi_c(x_{\partial c}).$$

Proof. By induction on the size of the subtree.

Base case: leaf node. Only one factor contributes, and the expression holds trivially.

Inductive step: assume the claim holds for all subtrees of depth $\leq k$. Then for a larger subtree,

$$Z_{a \rightarrow i}(x_i) = \sum_{x_{\partial a \setminus i}} \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} Z_{j \rightarrow a}(x_j),$$

which matches the factor-to-variable message update.

Thus the recursion exactly computes subtree marginals. $\square$

Max-Product Belief Propagation (MAP)

Replace sums by maxima.

- Variable $\rightarrow$ factor:

$$M_{i \rightarrow a}(x_i) = \prod_{b \in \partial i \setminus a} M_{b \rightarrow i}(x_i)$$

- Factor $\rightarrow$ variable:

$$M_{a \rightarrow i}(x_i) = \max_{x_{\partial a \setminus i}} \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} M_{j \rightarrow a}(x_j)$$

Remark 2.16.1. This computes max-marginals:

$$M_i(x_i) \propto \max_{x_{-i}} p(x)$$

2.16.1 Hidden Markov Models (HMMs)

Model

Latent chain:

$$z_1 \rightarrow z_2 \rightarrow \cdots \rightarrow z_n, \quad x_t | z_t$$

Factorization:

$$p(z_{1:n}, x_{1:n}) = p(z_1) \prod_{t=2}^n p(z_t | z_{t-1}) \prod_{t=1}^n p(x_t | z_t)$$

Algorithm 2 Max-product belief propagation

Initialize messages

repeat **for** factor a **do** update $M_{a \rightarrow i}$ **end for** **for** variable i **do** update $M_{i \rightarrow a}$ **end for****until** convergenceOutput $\arg \max_x p(x)$ from max-marginals

Forward Pass

Define

$$f_t(i) = p(z_t = i, x_{1:t})$$

Recursion:

$$f_{t+1}(j) = p(x_{t+1} | z_{t+1} = j) \sum_i f_t(i) p(z_{t+1} = j | z_t = i)$$

Backward Pass

$$b_t(i) = p(x_{t+1:n} | z_t = i)$$

Recursion:

$$b_t(i) = \sum_j p(z_{t+1} = j | z_t = i) p(x_{t+1} | z_{t+1} = j) b_{t+1}(j)$$

Posterior

$$p(z_t = i | x_{1:n}) \propto f_t(i) b_t(i)$$

Viterbi Algorithm (MAP)

Define

$$\delta_t(j) = \max_{z_{1:t-1}} p(z_{1:t}, x_{1:t})$$

Recursion:

$$\delta_{t+1}(j) = p(x_{t+1} | z_{t+1} = j) \max_i \delta_t(i) p(z_{t+1} = j | z_t = i)$$

Remark 2.16.2. Viterbi is max-product BP on a chain.

2.16.2 Loopy Belief Propagation

- Same updates as BP, but on graphs with cycles
- No guarantees of convergence or correctness
- Often works well in practice

Example 2.16.1. LDPC decoding uses loopy BP.

Summary: VE vs BP

- **Variable elimination:**
 - works on any graph
 - computes one marginal at a time
 - complexity depends on treewidth
- **Belief propagation:**
 - exact on trees
 - computes all marginals simultaneously
 - linear-time for trees
 - approximate on loopy graphs

2.17 Takeaways

- Exact inference is exponential in general
- Trees $\Rightarrow$ efficient dynamic programming
- Message passing unifies many algorithms:
 - forward-backward
 - Viterbi
 - BP
- MAP inference = max-product BP

2.18 Sampling with Markov Chains

Problem Formulation

We want to approximately sample from a distribution

$$p(x) \propto e^{f(x)}, \quad x \in \Omega,$$

within ε in total variation distance.

Remark 2.18.1. This problem appears across:

- Bayesian inference
- statistical physics
- combinatorics / theoretical CS
- deep generative modeling

Remark 2.18.2. Sampling is essentially equivalent to computing expectations:

$$\mathbb{E}_{x \sim p}[f(x)] = \int f(x)p(x) dx,$$

which becomes intractable in high dimensions.

2.18.1 Monte Carlo Estimation

Given samples $X_1, \dots, X_N \sim p$,

$$\mathbb{E}[f(X)] \approx \frac{1}{N} \sum_{i=1}^N f(X_i).$$

Proposition 2.18.1.

$$\text{Var}\left(\frac{1}{N} \sum_{i=1}^N f(X_i)\right) = \frac{\text{Var}(f(X))}{N}.$$

Proof. Uses independence:

$$\text{Var}\left(\frac{1}{N} \sum f(X_i)\right) = \frac{1}{N^2} \sum \text{Var}(f(X_i)) = \frac{1}{N} \text{Var}(f(X)).$$

□

What Distributions Are Easy to Sample?

- Uniform distributions
- Finite discrete distributions
- 1D distributions via inverse CDF
- Product distributions:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_i(x_i)$$

- Directed graphical models:

$$p(x) = \prod_{i=1}^n p(x_i | x_{\text{parents}(i)})$$

- Gaussian distributions

Remark 2.18.3. General high-dimensional distributions are not directly sampleable.

2.18.2 Importance Sampling

Let $Y \sim Q$.

Definition 2.18.1.

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \frac{p(Y_i)}{q(Y_i)} f(Y_i).$$

Proposition 2.18.2.

$$\mathbb{E}_Q[\hat{\mu}] = \mathbb{E}_P[f(X)].$$

Proof.

$$\mathbb{E}_Q \left[\frac{p(Y)}{q(Y)} f(Y) \right] = \int \frac{p(y)}{q(y)} f(y) q(y) dy = \int f(y) p(y) dy.$$

□

2.18.3 Rejection Sampling

Algorithm 3 Rejection sampling

Input: $q(x)$, c such that $p(x) \leq cq(x)$

repeat

 Sample $Y \sim Q$

 Sample $U \sim \text{Uniform}(0, 1)$

until $U \leq \frac{p(Y)}{cq(Y)}$

Output Y

Theorem 2.18.1. *If $p(x) \leq cq(x)$, then rejection sampling produces samples from p , and*

$$\# \text{trials} \sim \text{Geom}(1/c).$$

Remark 2.18.4. Fails in high dimensions due to exponential scaling of c .

2.19 Markov Chains

Definition 2.19.1. A sequence (X_t) is a Markov chain if

$$X_{t+1} \perp X_1, \dots, X_{t-1} \mid X_t,$$

i.e.,

$$p(x_{t+1} | x_1, \dots, x_t) = p(x_{t+1} | x_t).$$

Definition 2.19.2. A Markov chain is time-homogeneous if

$$p(x_{t+1} | x_t) \text{ does not depend on } t.$$

Transition Kernels

Definition 2.19.3. A Markov kernel K is a function such that:

- $K(x, \cdot)$ is a probability measure
- measurable in x

For a distribution μ ,

$$(\mu K)(A) = \int K(x, A) \mu(dx).$$

If densities exist:

$$(\mu K)(y) = \int \mu(x) k(x, y) dx.$$

Operator View

Definition 2.19.4. For $f : \Omega \rightarrow \mathbb{R}$,

$$Kf(x) = \mathbb{E}_{Y \sim K(x, \cdot)}[f(Y)] = \int f(y) K(x, dy).$$

Remark 2.19.1. K acts:

- on distributions: $\mu \mapsto \mu K$
- on functions: $f \mapsto Kf$

2.19.1 Stationary Distributions

Definition 2.19.5. π is stationary if

$$\pi K = \pi.$$

Remark 2.19.2. Then $\pi K^t = \pi$ for all t .

2.19.2 Conditions for Convergence

Definition 2.19.6. A Markov chain is:

- irreducible: any state can reach any other
- aperiodic: no deterministic cycles

Theorem 2.19.1. *If a finite Markov chain is irreducible and aperiodic:*

- *it has a unique stationary distribution π*
- $\mu_0 K^t \rightarrow \pi$

Theorem 2.19.2 (Ergodic theorem).

$$\frac{1}{N} \sum_{t=1}^N f(X_t) \rightarrow \mathbb{E}_\pi[f(X)] \quad a.s.$$

2.19.3 Reversibility

Definition 2.19.7. π satisfies detailed balance if

$$\pi(x)K(x, y) = \pi(y)K(y, x).$$

Proposition 2.19.1. *Detailed balance $\Rightarrow$ stationarity.*

Remark 2.19.3. Reversibility makes it easy to design Markov chains.

2.20 Metropolis–Hastings

Target density: $p(x) \propto \tilde{p}(x)$.

Algorithm 4 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x, y)$
for $t = 0, \dots, T$ **do**
 Sample proposal $\widehat{X}_{t+1} \sim q(X_t, \cdot)$
 Accept with probability

$$\alpha(x, y) = \min \left\{ \frac{\tilde{p}(y)q(y, x)}{\tilde{p}(x)q(x, y)}, 1 \right\}$$

end for

Theorem 2.20.1. *The MH chain has stationary distribution p .*

Gibbs Sampling

Definition 2.20.1. At each step:

$$X_i \sim p(x_i | x_{-i}).$$

Proposition 2.20.1. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability 1.*

Remark 2.20.1. In graphical models:

$$p(x_i | x_{-i}) = p(x_i | x_{\text{MB}(i)}).$$

Examples

Example 2.20.1 (Ball walk).

$$\widehat{X}_{t+1} \sim \text{Uniform}(B_R(X_t)),$$

then accept/reject via MH.

Example 2.20.2 (Gibbs sampling). Select coordinate i and sample

$$X_i \sim p(x_i | x_{-i}).$$

Takeaways

- Direct sampling is rarely possible
- MCMC constructs a Markov chain with target stationary distribution
- Convergence requires irreducibility + aperiodicity
- Reversibility simplifies construction (MH)
- Gibbs exploits conditional structure

2.21 Markov Chain Monte Carlo (MCMC)

2.21.1 Sampling via Markov Chains

Definition 2.21.1. Goal: sample from a distribution

$$p(x) \propto e^{f(x)}, \quad x \in \Omega,$$

within ε in total variation distance.

Remark 2.21.1. We construct a Markov chain whose stationary distribution is p , then simulate it.

Rejection Sampling (Limitations)

Theorem 2.21.1. If $p(x) \leq cq(x)$, rejection sampling produces samples from p , and

$$\# \text{trials} \sim \text{Geom}(1/c).$$

Remark 2.21.2. In high dimensions, c typically grows exponentially, making rejection sampling impractical.

Markov Chains

Definition 2.21.2. A sequence (X_t) is a Markov chain if

$$p(x_{t+1}|x_1, \dots, x_t) = p(x_{t+1}|x_t).$$

Definition 2.21.3. A distribution π is stationary if

$$\pi K = \pi.$$

2.21.2 Reversibility

Definition 2.21.4. π satisfies detailed balance if

$$\pi(x)K(x, y) = \pi(y)K(y, x).$$

Proposition 2.21.1. *If detailed balance holds, then π is stationary.*

Proof.

$$\sum_x \pi(x)K(x, y) = \sum_x \pi(y)K(y, x) = \pi(y) \sum_x K(y, x) = \pi(y).$$

□

Metropolis–Hastings

Algorithm 5 Metropolis–Hastings

Input: $\tilde{p}(x)$, proposal $q(x, y)$

for $t = 0, \dots, T$ **do**

 Sample $Y \sim q(X_t, \cdot)$

 Accept with probability

$$\alpha(x, y) = \min \left\{ \frac{\tilde{p}(y)q(y, x)}{\tilde{p}(x)q(x, y)}, 1 \right\}$$

 Set $X_{t+1} = Y$ if accepted, else $X_{t+1} = X_t$

end for

Theorem 2.21.2. *The Metropolis–Hastings chain has stationary distribution p .*

Proof. Check detailed balance:

$$p(x)K(x, y) = p(x)q(x, y)\alpha(x, y) = \min\{p(x)q(x, y), p(y)q(y, x)\},$$

which is symmetric in (x, y) .

□

Gibbs Sampling

Definition 2.21.5. At each step, pick coordinate i and sample

$$X_i \sim p(x_i | x_{-i}).$$

Proposition 2.21.2. *Gibbs sampling is a special case of Metropolis–Hastings with acceptance probability 1.*

Remark 2.21.3. In graphical models:

$$p(x_i | x_{-i}) = p(x_i | x_{\text{MB}(i)}).$$

2.22 Langevin Monte Carlo (LMC)

Continuous-time process

Definition 2.22.1. Langevin diffusion:

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t.$$

Remark 2.22.1. Target distribution:

$$p(x) \propto e^{-V(x)}.$$

Discretization

Using Euler–Maruyama:

$$X_{t+1} = X_t - \eta \nabla V(X_t) + \sqrt{2\eta} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I).$$

Remark 2.22.2. This is noisy gradient descent.

Stationary Distribution

Theorem 2.22.1. *The Langevin diffusion has stationary distribution*

$$p(x) \propto e^{-V(x)}.$$

Proof. From the Fokker–Planck equation:

$$\frac{\partial \mu_t}{\partial t} = \nabla \cdot (\nabla V \mu_t) + \Delta \mu_t.$$

Plug $\mu(x) = e^{-V(x)}$:

$$\nabla \mu = -(\nabla V) e^{-V}, \quad \Delta \mu = (-\Delta V + \|\nabla V\|^2) e^{-V}.$$

Then:

$$\nabla \cdot (\nabla V \mu) = (\Delta V) e^{-V} - \|\nabla V\|^2 e^{-V}.$$

Adding:

$$\nabla \cdot (\nabla V \mu) + \Delta \mu = 0.$$

Thus μ is stationary. □

2.22.1 MALA (Metropolis-adjusted Langevin)

Remark 2.22.3. Corrects discretization bias of LMC.

Algorithm 6 MALA

Propose

$$Y \sim \mathcal{N}(x - h\nabla V(x), 2hI)$$

Accept with MH correction

2.23 Underdamped Langevin Dynamics

Definition 2.23.1. Introduce momentum v :

$$\begin{aligned} dX_t &= v_t dt \\ dv_t &= -\gamma v_t dt - \nabla V(X_t) dt + \sqrt{2\gamma} dB_t \end{aligned}$$

Remark 2.23.1. Stationary distribution:

$$p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}.$$

Remark 2.23.2. Non-reversible $\rightarrow$ faster mixing.

Hamiltonian Monte Carlo (HMC)

Hamiltonian dynamics

Define:

$$H(q, p) = U(q) + K(p)$$

$$\begin{aligned} \frac{dq}{dt} &= \nabla_p H = \nabla K(p) \\ \frac{dp}{dt} &= -\nabla_q H = -\nabla U(q) \end{aligned}$$

Proposition 2.23.1. *Hamiltonian dynamics preserve:*

- *Energy:* $H(q(t), p(t))$
- *Volume (Liouville's theorem)*

Reversibility trick

Define momentum flip:

$$R(q, p) = (q, -p).$$

Then:

$$R \circ S_T \circ R = S_T^{-1}.$$

Proposition 2.23.2. *HMC transition is reversible w.r.t.*

$$\mu(q, p) \propto e^{-H(q, p)}.$$

Algorithm 7 Hamiltonian Monte Carlo (idealized)

Sample $p \sim e^{-K(p)}$ Simulate Hamiltonian dynamics for time T Output q

Algorithm**Remark 2.23.3.** No rejection needed in exact dynamics (but required after discretization).**Leapfrog Integrator**

Algorithm 8 Leapfrog

 $p \leftarrow p - \frac{h}{2} \nabla U(q)$ **for** $t = 1, \dots, T$ **do** $q \leftarrow q + h \nabla K(p)$ $p \leftarrow p - h \nabla U(q)$ **end for** $p \leftarrow p - \frac{h}{2} \nabla U(q)$

Remark 2.23.4. Properties:

- volume-preserving
- reversible
- approximately energy-preserving

Takeaways

- MCMC constructs Markov chains with target stationary distribution
- Metropolis–Hastings ensures correctness via detailed balance
- Gibbs exploits conditional structure
- Langevin adds gradient information (diffusion-based sampling)
- HMC uses Hamiltonian dynamics for efficient exploration

Langevin Monte Carlo

We want to sample from

$$p(x) \propto e^{-V(x)}.$$

Definition 2.23.2. The Langevin diffusion is the stochastic differential equation

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dB_t.$$

Discretizing via Euler–Maruyama:

$$X_{t+1} = X_t - \eta \nabla V(X_t) + \sqrt{2\eta} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d).$$

Remark 2.23.5. This can be viewed as gradient descent with injected Gaussian noise.

Remark 2.23.6. The continuous-time process has stationary distribution $p(x) \propto e^{-V(x)}$, but the discretized version introduces bias unless η is small.

Metropolis-Adjusted Langevin Algorithm (MALA)

To correct discretization bias, use Langevin as a proposal in Metropolis–Hastings.

Definition 2.23.3. Proposal distribution:

$$q(x, y) \propto \exp\left(-\frac{\|y - (x - \eta \nabla V(x))\|^2}{4\eta}\right).$$

Algorithm 9 MALA

Propose $Y \sim \mathcal{N}(x - \eta \nabla V(x), 2\eta I)$

Accept with probability

$$\alpha(x, y) = \min\left\{\frac{p(y)q(y, x)}{p(x)q(x, y)}, 1\right\}$$

Underdamped Langevin Dynamics

Definition 2.23.4. Introduce momentum v :

$$\begin{aligned} dX_t &= v_t dt, \\ dv_t &= -\gamma v_t dt - \nabla V(X_t) dt + \sqrt{2\gamma} dB_t. \end{aligned}$$

Remark 2.23.7. Stationary distribution:

$$p(x, v) \propto e^{-V(x) - \frac{1}{2}\|v\|^2}.$$

Remark 2.23.8. Noise is injected only in v_t , leading to more efficient exploration.

Continuous-Time Markov Processes

Definition 2.23.5. A continuous-time Markov process $(X_t)_{t \geq 0}$ satisfies

$$P(X_u \in A | X_{[0,t]}) = P(X_u \in A | X_t), \quad u > t.$$

Definition 2.23.6. A family $(P_t)_{t \geq 0}$ is a Markov semigroup if

$$P_{t+s} = P_t P_s, \quad P_0 = \text{id}.$$

Generator of a Markov Process

Definition 2.23.7. The generator $\mathcal{L}$ is

$$\mathcal{L}f = \left. \frac{d}{dt} P_t f \right|_{t=0}.$$

Remark 2.23.9. This defines the process via

$$P_t = e^{t\mathcal{L}}.$$

2.23.1 Fokker–Planck Equation

Let $\mathcal{L}^*$ be the adjoint operator.

Definition 2.23.8. The distribution μ_t evolves as

$$\frac{d\mu_t}{dt} = \mathcal{L}^* \mu_t.$$

Remark 2.23.10. For Langevin diffusion:

$$\mathcal{L}f = -\langle \nabla f, \nabla V \rangle + \Delta f,$$

$$\mathcal{L}^* \mu = \nabla \cdot (\nabla V \mu) + \Delta \mu.$$

Stationarity of Langevin

Theorem 2.23.1. *The distribution*

$$\mu(x) \propto e^{-V(x)}$$

is stationary for Langevin diffusion.

Proof. Plug into Fokker–Planck equation:

$$\mathcal{L}^* \mu = \nabla \cdot (\nabla V \mu) + \Delta \mu.$$

Compute:

$$\nabla \mu = -(\nabla V) \mu, \quad \Delta \mu = (-\Delta V + \|\nabla V\|^2) \mu.$$

Then:

$$\nabla \cdot (\nabla V \mu) = (\Delta V) \mu - \|\nabla V\|^2 \mu.$$

Adding:

$$(\Delta V - \|\nabla V\|^2) \mu + (-\Delta V + \|\nabla V\|^2) \mu = 0.$$

□

2.24 Hamiltonian Monte Carlo (HMC)

2.24.1 Hamiltonian Dynamics

Define

$$H(q, p) = U(q) + K(p).$$

Dynamics:

$$\frac{dq}{dt} = \nabla_p H, \quad \frac{dp}{dt} = -\nabla_q H.$$

Remark 2.24.1. Typical choice:

$$K(p) = \frac{1}{2}\|p\|^2.$$

2.24.2 Key Properties

Proposition 2.24.1. *Hamiltonian dynamics preserve:*

- *energy $H(q, p)$*
- *volume in phase space*

2.24.3 Leapfrog Integrator

Lemma 2.24.1. *The leapfrog integrator:*

- *preserves volume*
- *is reversible (up to momentum flip)*

Remark 2.24.2. It does *not* exactly preserve the Hamiltonian.

HMC with Leapfrog

Algorithm 10 Hamiltonian Monte Carlo (practical)

Sample $p \sim e^{-K(p)}$

Simulate Hamiltonian dynamics via leapfrog for time T

Accept with probability

$$\alpha = \min\{e^{-H(q', p') + H(q, p)}, 1\}$$

Remark 2.24.3. The Metropolis step corrects discretization error.

Choosing Parameters

- Integration time T :
 - Too small $\Rightarrow$ slow exploration
 - Too large $\Rightarrow$ wasted computation
- Randomizing T helps avoid periodic behavior
- No-U-Turn criterion adaptively selects trajectory length

Geometric Adaptation

Definition 2.24.1. Riemannian HMC uses position-dependent metric:

$$K(q, p) = \frac{1}{2} p^\top \Sigma^{-1}(q) p + \frac{1}{2} \log \det \Sigma(q).$$

2.25 Convergence of Markov Chains

2.25.1 Spectral Gap

Theorem 2.25.1. Let λ_2 be the second largest eigenvalue. Then

$$\text{Var}_\mu(K^n f) \leq |\lambda_2|^{2n} \text{Var}_\mu(f).$$

2.25.2 Non-reversible Chains

Remark 2.25.1. For non-reversible chains:

$$\chi^2(\nu K^n || \mu) \leq C |\lambda_2|^{2n} \chi^2(\nu || \mu).$$

Remark 2.25.2. In continuous time this is called *hypocoercivity*.

2.25.2.1 Poincaré Inequality

Definition 2.25.1. A distribution μ satisfies a Poincaré inequality if

$$\langle g, \mathcal{L}g \rangle_\mu \geq c \text{Var}_\mu(g).$$

Remark 2.25.3. Implies exponential convergence rate e^{-2ct} .

2.25.3 Contraction in Divergences

Definition 2.25.2. A kernel K is contractive if

$$D_f(\nu K || \mu) \leq (1 - \kappa) D_f(\nu || \mu).$$

Remark 2.25.4. Implies geometric convergence.

2.25.4 Mixing Time

Definition 2.25.3.

$$t_{\text{mix}}(\varepsilon) = \min\{t : \sup_{\mu_0} \text{TV}(\mu_0 K^t, \mu) \leq \varepsilon\}.$$

Definition 2.25.4. Relaxation time:

$$t_{\text{rel}} = \frac{1}{1 - |\lambda_2|}.$$

2.25.5 Conductance

Definition 2.25.5. For $S \subset \Omega$:

$$\Phi(S) = \frac{Q(S, S^c)}{\mu(S)}, \quad Q(S, S^c) = \sum_{x \in S, y \in S^c} \mu(x) P(x, y).$$

Definition 2.25.6.

$$\Phi_* = \min_{\mu(S) \leq 1/2} \Phi(S).$$

Theorem 2.25.2.

$$t_{\text{mix}}(1/4) \geq \frac{1}{4\Phi_*}.$$

2.25.6 Asymptotic Variance

Definition 2.25.7.

$$\sigma_f^2 = \text{Var}_\mu(f(X_0)) + 2 \sum_{k=1}^{\infty} \text{Cov}_\mu(f(X_0), f(X_k)).$$

Theorem 2.25.3.

$$\frac{1}{N} \sum_{k=0}^{N-1} f(X_k) \rightarrow \mathbb{E}_\mu[f]$$

and

$$\text{Var}\left(\frac{1}{N} \sum f(X_k)\right) \sim \frac{\sigma_f^2}{N}.$$

2.25.7 Practical Diagnostics

- Trace plots (check stationarity)
- Autocorrelation:

$$\rho_\ell = \frac{\text{Cov}(f(X_0), f(X_\ell))}{\text{Var}(f)}$$

- Effective sample size:

$$N_{\text{eff}} = \frac{N}{1 + 2 \sum_{\ell \geq 1} \rho_\ell}$$

Discussion

- Single long chain vs multiple shorter chains:
 - single chain: cheaper burn-in
 - multiple chains: better exploration of multimodal distributions

2.26 Tempering and Particle Methods

2.26.1 Multimodality and Slow Mixing

Multimodal target distributions pose a fundamental challenge for MCMC.

- Probability mass is concentrated in separated regions (modes).
- Transitions between modes require crossing *energy barriers*.
- Local proposals (e.g., random walk, Langevin) rarely cross these barriers.

Remark 2.26.1. For distributions of the form $p(x) \propto e^{-f(x)}$, the time to transition between modes is often *exponential* in the barrier height:

$$\text{mixing time} \sim \exp(\Delta f).$$

This is known as *metastability*.

2.26.2 Temperature Scaling

Introduce a *temperature parameter* $T > 0$:

$$p_T(x) \propto e^{-f(x)/T} = e^{-\beta f(x)}, \quad \beta = \frac{1}{T}.$$

- High temperature ($T \gg 1$, $\beta \ll 1$): distribution is *flattened*.
- Low temperature ($T \ll 1$, $\beta \gg 1$): distribution concentrates near modes.

Remark 2.26.2. In Langevin dynamics:

$$dX_t = -\nabla V(X_t) dt + \sqrt{2T} dB_t,$$

temperature controls the tradeoff between gradient (drift) and noise.

2.26.3 Simulated Annealing

Simulated annealing gradually lowers temperature:

$$T_1 > T_2 > \dots \rightarrow 0.$$

- Initially explores broadly.
- Eventually concentrates near global minima.

Remark 2.26.3. Once modes separate at low temperature, the chain becomes *trapped*: probability mass cannot easily move between modes.

2.26.4 Simulated Tempering

Introduce temperature as an auxiliary variable:

$$(\ell, x) \in \{1, \dots, L\} \times \Omega.$$

Define target:

$$\pi(\ell, x) \propto w_\ell p_\ell(x), \quad p_\ell(x) \propto e^{-f(x)/T_\ell}.$$

Algorithm:

- Alternate:
 - Update x using MCMC at fixed temperature ℓ .
 - Propose $\ell \rightarrow \ell \pm 1$ with MH acceptance:

$$\alpha = \min \left\{ \frac{w_{\ell'} p_{\ell'}(x)}{w_\ell p_\ell(x)}, 1 \right\}.$$

Remark 2.26.4. Drawback: requires knowledge (or estimation) of normalizing constants for p_ℓ .

2.26.5 Parallel Tempering (Replica Exchange)

Run multiple chains at different temperatures:

$$(x_1, \dots, x_L), \quad x_\ell \sim p_\ell.$$

Target:

$$\pi(x_1, \dots, x_L) = \prod_{\ell=1}^L p_\ell(x_\ell).$$

Swap proposal:

$$(x_i, x_{i+1}) \rightarrow (x_{i+1}, x_i),$$

with acceptance probability

$$\alpha = \min \left\{ \frac{p_i(x_{i+1}) p_{i+1}(x_i)}{p_i(x_i) p_{i+1}(x_{i+1})}, 1 \right\}.$$

Remark 2.26.5. Advantages:

- No need for normalizing constants.
- High-temperature chains help low-temperature chains escape modes.

2.26.6 Sequential Monte Carlo (SMC)

Construct a sequence of distributions:

$$p_k(x) \propto \tilde{p}_k(x), \quad k = 1, \dots, L,$$

typically interpolating between easy and target distributions.

Algorithm 11 Sequential Monte Carlo

- 1: Sample $X_1^{(i)} \sim p_1$
- 2: **for** $k = 2$ to L **do**
- 3: Compute weights:

$$w_k^{(i)} \propto \frac{\tilde{p}_k(X_{k-1}^{(i)})}{\tilde{p}_{k-1}(X_{k-1}^{(i)})}$$

- 4: Resample particles according to $w_k^{(i)}$
 - 5: Move particles: $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
 - 6: **end for**
 - 7: Output $\{X_L^{(i)}\}$
-

Remark 2.26.6. SMC combines:

- Importance sampling (weights)
- Resampling (to prevent degeneracy)
- MCMC moves (to diversify particles)

2.26.7 Annealed Importance Sampling (AIS)

A simplified version of SMC without resampling.

Algorithm 12 Annealed Importance Sampling

- 1: Sample $X_1^{(i)} \sim p_1$
- 2: **for** $k = 2$ to L **do**
- 3: Sample $X_k^{(i)} \sim K_k(X_{k-1}^{(i)}, \cdot)$
- 4: **end for**
- 5: Compute weights:

$$\tilde{w}^{(i)} = \prod_{k=2}^L \frac{\tilde{p}_k(X_k^{(i)})}{\tilde{p}_{k-1}(X_k^{(i)})}$$

- 6: Estimate:

$$\mathbb{E}_{p_L}[f(X)] \approx \frac{\sum_i \tilde{w}^{(i)} f(X_L^{(i)})}{\sum_i \tilde{w}^{(i)}}$$

Remark 2.26.7. AIS is especially useful for:

- Estimating partition functions
- Bridging between simple and complex distributions

Key Insight

Tempering and particle methods address the same core issue:

Exploration vs. concentration tradeoff

- High temperature $\Rightarrow$ exploration
- Low temperature $\Rightarrow$ accurate sampling
- Tempering methods: move between temperatures
- Particle methods: propagate distributions across temperatures